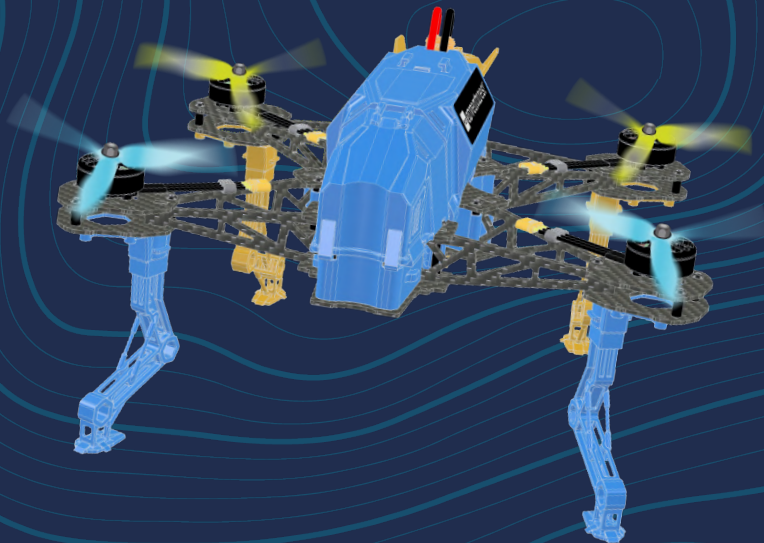


Drone Pathways

Foundations of Flight Operations



GRADES 10-12

Modular Platform



Build



Curriculum Tracks



Simulate



Mission Planning



Fly



CURRICULUM SAMPLE



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(208) 343-3110



FLEX UAV



COMPLETE KIT



FLIGHT SIMULATOR



Drone Pathways

Foundations of Flight Operations

GRADES: 10-12

STUDENTS Varies by package	TIME Twenty, 45-minute lessons
SUBJECTS Technology	SETTINGS - Classrooms - CTE

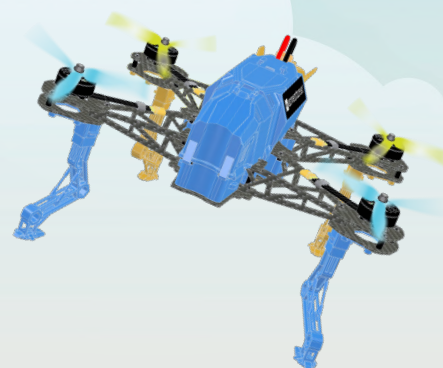
Aligned to CTE standards, *Drone Pathways* transforms classrooms into aviation hubs. By learning to operate as a flight crew, students master essential **teamwork** and **collaboration**.

TECH REQUIREMENTS

- One Internet-connected device per learner (web-based content is compatible with Chromebooks, Windows and Mac)

PRICING OPTIONS

- CTE Starter Package - 2 Drones: \$3,995⁰⁰
- CTE Base Package - 5 Drones: \$8,995⁰⁰
- CTE Plus Package - 10 Drones: \$16,995⁰⁰
- CTE Pro Package - 10 Drones: \$22,995⁰⁰



SCAN OR CLICK QR CODE FOR:

PRODUCT ORIENTATION

FULL MATERIAL LIST

STANDARDS & ALIGNMENT

CONTACT US:

Call: (208) 343-3110

Email: sales@edventures.com

Web: edventures.com



Drone Pathways

Foundations of Flight Operations



Instructor Guide



CURRICULUM OVERVIEW

The Drone Pathways curriculum is designed as a comprehensive career and technical education (CTE) workflow, moving students from the workbench to the pilot's seat. Unlike simple "flying" programs, this curriculum emphasizes the systems, safety and soft skills (Crew Resource Management) required in the professional drone industry.

INSTRUCTIONAL APPROACH

Scenario-Based Learning: Throughout the curriculum, students engage in narrative-driven missions, such as inspecting a downed power line or surveying sensitive wetlands. These scenarios are designed to bridge the gap between abstract technical skills and real-world application, giving students a "why" for the skills they are learning.

- **Note on realism:** Be aware that these scenarios are crafted primarily as thought experiments and educational simulations. While they are rooted in real industry applications, some details may be dramatized or simplified to maximize student engagement and fit within a 45-minute class period. They should be viewed as prompts to spark critical thinking and discussion rather than precise preparation for industry-specific operations, which require advanced technical training beyond the scope of this introductory course.

Classroom-Adapted Team Dynamics: In the professional drone industry, a flight crew in the field often consists of a Pilot and one Visual Observer. To fit the classroom model and ensure every learner is engaged, Drone Pathways adapts these structures into larger 3–5 person teams.

- **Why we do this:** This structure moves beyond the "one person flies, everyone else watches" trap. By expanding the crew to include specialized roles, we prioritize durable skills over niche technical abilities. While manual piloting is specific to aviation, the human skills of Crew Resource Management—clear communication, situational awareness and accountability—are universal assets. This gives every student the opportunity to gain experience relevant to any career path.

CURRICULUM BREAKDOWN

The Drone Pathways program consists of 20 core lessons, culminating in a Capstone Mission. While the lessons follow a linear progression, Build → Simulate → Fly, the pacing is flexible to accommodate different learning environments.

The curriculum is broken down into the following three sections:

- **Section 1: Hardware—UAV Fundamentals:** This section focuses on the physical assembly of the modular FLEX UAV. Students learn to identify components, study essential safety protocols like LiPo battery handling and understand CRM principles.
 - **Why this matters:** Students don't just open a box; they learn mechanical assembly and the function of subsystems (structural, mechanical, electrical). This phase builds "ownership" of the drone. If they crash it later, they know how to fix it because they built it.
- **Section 2: Software—Flight Sim Skills:** Students transition to hands-on training using the flight simulator to develop essential piloting proficiency in a risk-free virtual environment. They master the four primary radio controls—pitch, roll, yaw and throttle—by completing structured tutorials and gamified practice scenarios.
 - **Why this matters:** This is the risk-free training ground. Students must master throttle, pitch, roll, and yaw and demonstrate proficiency in complex maneuvers (like the figure eight) before they are cleared for live equipment. This phase also introduces "Mission Planning," moving students from "gamers" to "operators."

- **Section 3: Flight—Regulations & Piloting:** This final phase applies learned skills to live flight, emphasizing FAA compliance (including the TRUST certification) and mastering precise visual line-of-sight (VLOS) operations and pre-flight safety checks in preparation for capstone missions.
 - Why this matters: This phase applies the simulation skills to the real world. It covers legal requirements (FAA Part 107 vs. Recreational), ethical decision-making and the execution of the Capstone Mission.

Extended Curriculum: By utilizing the Extensions & Explore More activities found at the end of each lesson, you can significantly expand the program. These include deep dives into GIS careers, custom CAD design challenges for landing gear and advanced flight maneuver drills.

EFFECTIVE LESSON USE

To get the most out of the Drone Pathways curriculum, it is helpful to understand the structure of the lesson plans and how to adapt them to your specific learning environment.

Preparation Phase (Pre-Class Readiness):

- Objectives and Time Required: These elements clearly define the target learning outcomes (install landing gear on the FLEX UAV) and the expected 45-minute duration for the session. Use this to align the lesson with your overall program schedule and pacing guide.
- Prep and Materials: This section serves as the critical safety and logistics checklist. It reminds you to verify both physical items (drone kits, Pilot Logs, safety goggles) and digital access (accompanying slides, build guides, flight simulator) are ready before students enter the class.
- Safety Precautions: This is a mandatory pre-class review, confirming essential rules such as “Keep LiPo batteries stored in the fireproof container” and “Keep props off.”
- Key Terms: Provides a quick reference for the precise subject terminology you must understand before defining concepts for the students.

Execution Phase (Instruction & Adaptation):

- Schedule and Step-by-Step Instructions: These sections provide the detailed time blocks and logical sequence for executing the lesson. The structure allows you to follow the sequence directly, minimizing on-the-fly decision-making
- CRM Integration: The instructions guide you to embed CRM techniques (verbalization, role rotation, assertiveness) directly into technical tasks, such as encouraging students to assign roles for building steps.
 - Implementation Strategy: Treat the classroom like a job site. Encourage students to take roles seriously. The Visual Observer isn't just watching—they are an active safety officer. This focus on soft skills turns a fun activity into workforce preparation.

Adapting for Your Learning Environment:

For out-of-school, enrichment or drone clubs, if time is limited, you may condense the theoretical discussions to focus heavily on the build and flight mechanics. However, never skip the safety or simulator lessons, as these are non-negotiable barriers to safe flight operations.

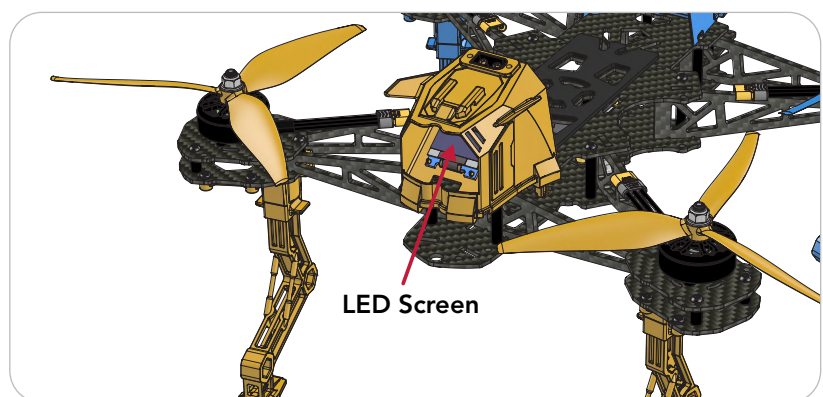
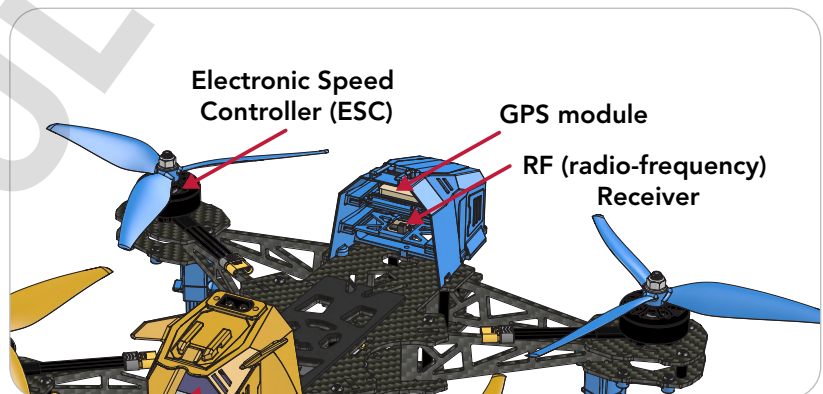
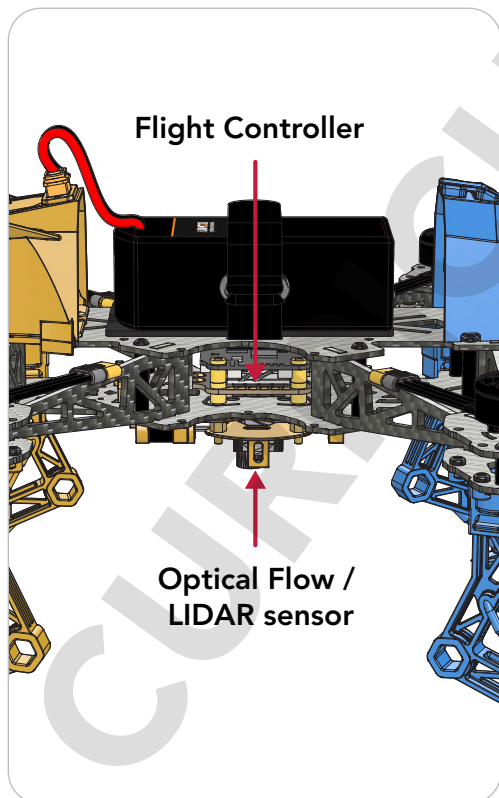
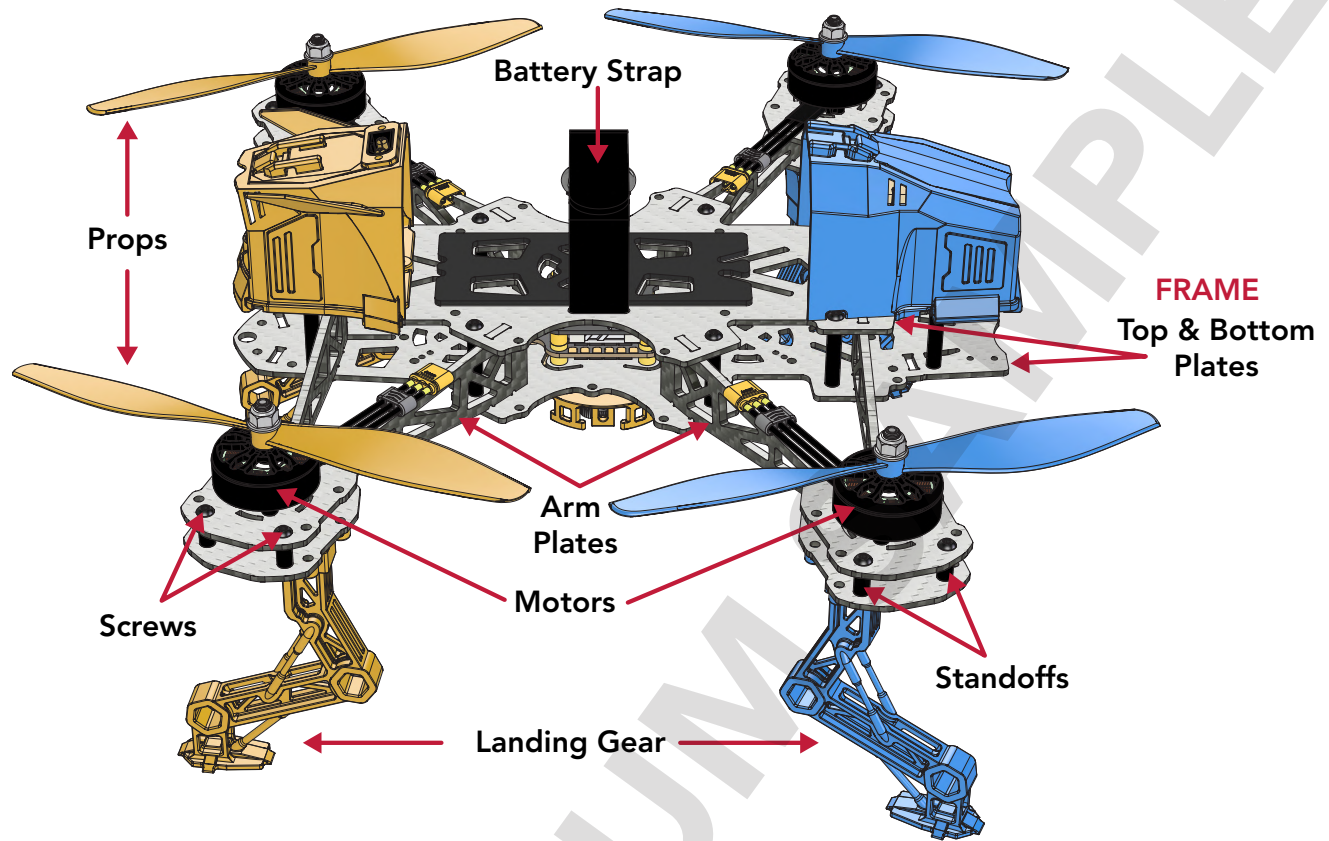
Review and Differentiation:

- Assessments: The formative and summative assessments provide clear, built-in methods for evaluating student learning immediately (formative) and measuring mastery of the lesson objectives (summative). This allows you to gather data and adjust the depth of the next lesson based on student performance.
- Extensions and Explore More: This section offers specific ideas for building on the learning objectives aligned to the lesson, providing immediate, relevant work for advanced students who finish early or suggesting in-depth projects that extend the core concepts.

- For Advanced Students: Consider using these prompts to provide relevant challenges while others finish their activities.
- For Larger Classes: Use these activities to keep half the class engaged in research or design while the other half completes objectives in the Flight Simulator.

CURRICULUM SAMPLE

Lead a discussion on the function and importance of key components, including the frame, arms, motors, propellers, landing gear, sensors and flight controller.



Lesson 12:

Mission Planning & Flight Team Operations

TIME REQUIRED: 45 minutes



OVERVIEW

Plan a detailed drone flight mission, incorporating Crew Resource Management (CRM) principles for effective flight team operations and reinforcing core piloting skills.



SCHEDULE

Topics:

- ☐ Principles of Drone Mission Planning (10 min.)
- ☐ Create a Flight Mission Plan (15 min.)
- ☐ Flight Sim Practice (10 min.)
- ☐ Pilot Log Entry & Wrap Up (10 min.)



MATERIALS

- Slides
- Pilot Logs (1 per student)
- Internet connected devices with Google Earth (or similar tool) and Flight Sim access (1 per student)
- Flight Sim Controller (1 per student)



SAFETY PRECAUTIONS

- Keep props off.
- Keep LiPo batteries stored in the fireproof container.



OBJECTIVES

- Identify parts of a flight mission plan.
- Use Google Earth or a similar GIS application to create a basic flight mission plan.
- Continue to practice on the flight sim.



KEY TERMS

- **Drone mission planning:** The process of preparing for a flight, which includes defining clear objectives, careful route planning and considering safety, efficiency and team roles.
- **Flight mission plan:** The documented output of the planning process, which for this lesson includes a location pin, imagined hazard (a path for a downed power line), safety zones (polygons for exclusion zones) and a flight path plotted on a digital map.
- **Geographic information system (GIS):** A computer system that captures, stores, checks, and displays data related to positions on Earth's surface.

STEP-BY-STEP INSTRUCTIONS



Whole Group

Principles of Drone Mission Planning

Explain the fundamental principles of drone mission planning, emphasizing that successful operations require clear objectives, careful route planning and consideration of factors like safety, efficiency and well-defined team roles.

Have students brainstorm various types of drone-based missions (delivery, inspection, data collection for specific industries, search and rescue) and how planning differs for each. Highlight the importance of a well-defined mission and a thorough pre-mission briefing (a key CRM element) for the entire crew before any flight. Have students recall the downed power-line scenario from Lesson 1 as an example:

Downed Power-Line Pre-Mission Briefing Example

Mission:

- Inspect the downed live power line and the surrounding area.
- Gather critical data for emergency management.
- Assess damage and fire hazard.

Key Risks:

- Live power line
- Extreme fire risk (dry conditions)
- Uninformed local population
- Weather conditions
- Wildlife

Team Roles & Comms:

- RPIC: Fly drone, execute maneuvers, practices closed loop communication.
- VO: Monitor hazards (line, fire), call out conditions (e.g., "Wire clear," "Smoke sighted").
- Data Manager/Safety: Ensure data capture, monitor ground safety.
- Communication: Clear, concise, closed-loop. Always confirm important calls.

Contingency:

- Lost VLOS: Immediate return to home (RTH).
- Fly Away Procedure: Immediate RTH.
- Drone Malfunction: Controlled landing in safe zone.
- Fire Ignition: Abort mission, notify emergency services.



Small Groups

Create a Flight Mission Plan



Have student teams open Google Earth (or a similar mapping program) and choose a familiar, relatively open location near them, such as their school's football field, a large park or a nearby open field. Instruct them to imagine that a critical power line has fallen across or adjacent to this location and the conditions are hot and dry.

Remind students that this is one of the most hazardous situations a drone operator could face, and those who choose this line of utility inspection work undertake hundreds of hours of specialized training to be prepared for these advanced missions. While they have not yet received this training and should not attempt this kind of mission without it, the example highlights the importance of strong CRM practices for successful mission planning. While this lesson doesn't cover all the specific safety protocols that would need to be in place to mitigate the risk of this type of mission, learners should focus on applying the general CRM practices that they've learned so far.

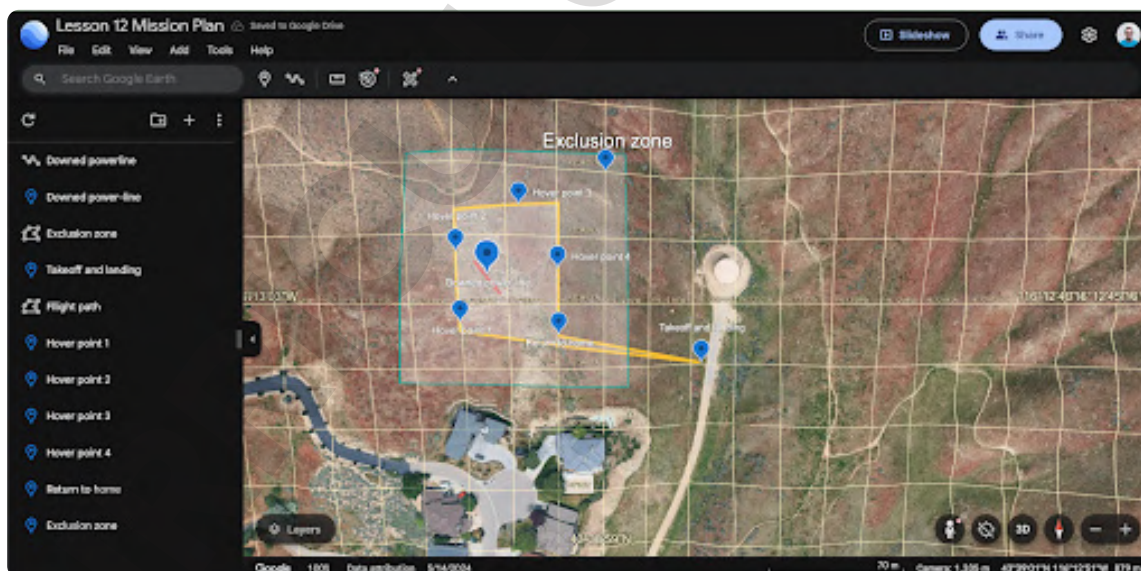
Downed Power-Line Mission Example

Mission Objective:

- Simulate inspecting an imagined downed power line and surrounding area at the chosen location.
- Assess imagined damage and identify simulated fire hazards for emergency services.

Google Earth Planning:

- Location pin: Pinpoint your chosen location (your school's football field) on Google Earth.
- Imagined hazard: Use the "Add Path" tool to draw an imagined downed power line across or along the edge of your chosen field.
- Safety zones: Use the "Add Polygon" tool to mark imagined exclusion zones around the power line where human crews cannot go due to safety risks.
- Flight path: Plot a flight path over the area where the power line is imagined, ensuring coverage for inspection.



Flight Plan Details:

- Takeoff/landing zone: Designate a clear, safe area on the field as your takeoff/landing zone.
- Altitude: Maintain 75 ft. AGL (above ground level) for inspection grid, adjusting for any imagined tall structures.
- Maneuvers: Execute a slow, steady pass in a grid with defined “hover points” for inspection. Use precise pitch/roll for accurate coverage over the imagined downed line.
- Ground features: Identify and note key ground features like dry grass, adjacent buildings or access roads via Google Earth’s satellite imagery.

Team Roles & Communication:

- RPIC: Controls drone, executes planned grid pattern over the imagined power line. Practices closed-loop communication.
- VO: Monitors drone and ground, calls out imagined hazards (e.g., “Imagined wire clear!” “Dry patch at 3 o’clock!”).
- Communication: “Starting grid over power line,” “Confirming imagery capture,” “Spotting potential fire risk.”

Contingency:

- If drone malfunctions: RPIC initiates immediate Return-to-Home.
- If imagined fire ignites: Abort mission and initiate immediate Return-to-Home, notify emergency services.

**Individual****Flight Sim Practice**

Direct students to practice on the flight simulator as they finish their mission plans.

- Target: Practice in the FLEX UAV Flight Simulator, focusing on checking off any remaining items in the FLEX UAV Flight Simulator Checklist or beating their best time.



- Logging: Instruct students to make a quick note in their Pilot Log about this practice flight, focusing on their precision and control.

**Individual****Pilot Log Entry & Wrap Up**

Have students get their Pilot Logs out and reflect on how precise planning, proactive risk assessment and effective team communication (CRM) connect to successful outcomes in real-world drone-related careers.



ASSESSMENTS

- **Formative Assessment:** “Mission Plan Checklist & Peer Feedback.” Have student teams review each other’s drafted mission plans, focusing on completeness, logical flow and feasibility. They can provide brief verbal feedback to their peers, assessing their ability to identify strengths and weaknesses in a plan, especially concerning CRM aspects like defined roles and communication.
- **Summative Assessment:** “Mission Plan Presentation.” Student teams present their mission plan (e.g., screenshots of Google Earth/Maps labeled with mission route and objectives) and how CRM principles (defined roles, planned communication, risk mitigation, contingency planning for advanced scenarios) were integrated into their plan.

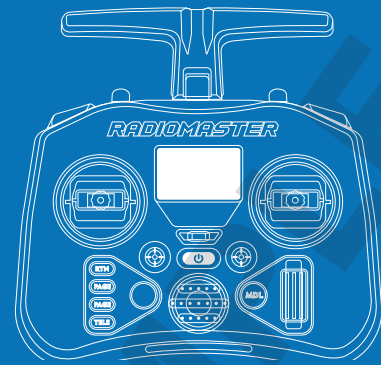


EXTENSIONS & EXPLORE MORE

- **Power Line Inspection Safety Research:** Have students research the traditional method of inspecting high-voltage power lines. (Answer: Highly trained “linemen” must physically climb poles or use bucket trucks, facing risks of falls and electrocution). Challenge students to investigate how utility companies are now using drones for inspection. Have them find a video or article demonstrating a drone inspection and list three specific defects the pilot is looking for.
- **Drones Applications in Science:** Have students research how drones are currently being used in science fields and choose one application they find interesting (e.g., “Monitoring Volcanoes” or “Tracking Polar Bear Populations”). Students must find out how this specific task was performed before drones existed. Was it done by satellite? Helicopter? Hiking on foot? Have students identify exactly how the drone improved the process. Did it improve safety (removing humans from danger), cost (cheaper than a helicopter) and/or data quality (closer/higher resolution)?
- **GIS Career Path Deep Dive:** Building on the foundational use of Google Earth (a basic GIS application), challenge students to research the field of Geographic Information Systems (GIS) and the professional role of a geospatial analyst or GIS specialist in drone-related careers (environmental monitoring, land surveying, infrastructure inspection). They should create a job profile outlining some necessary skills and explaining how data collected by drones informs their decisions.
- **Airspace Authorization Research:** Building on the mission planning exercise, have students research the process of obtaining airspace authorization using a Low Altitude Authorization and Notification Capability (LAANC) service provider. Challenge them to identify the airspace classification of a local airport or controlled zone, determine if authorization is required to fly nearby and list the steps necessary for legal flight within that airspace.
- **Geospatial Analysis:** Have students perform a simple Land Use/Land Cover (LULC) Change Analysis.
 1. Select a rapidly developing or changing local area
 2. Use the “Historical Imagery” slider in Google Earth to view the area in two distinct years (10 or 20 years apart)
 3. Use the “Add Polygon” tool to define and map the boundary of one specific land type (an area of forest, new residential construction or open field) for both time periods.
 4. Calculate the difference in area to quantify the change, demonstrating how GIS tracks development and environmental impact over time.

Radio Controller Guide for

FLEX UAV



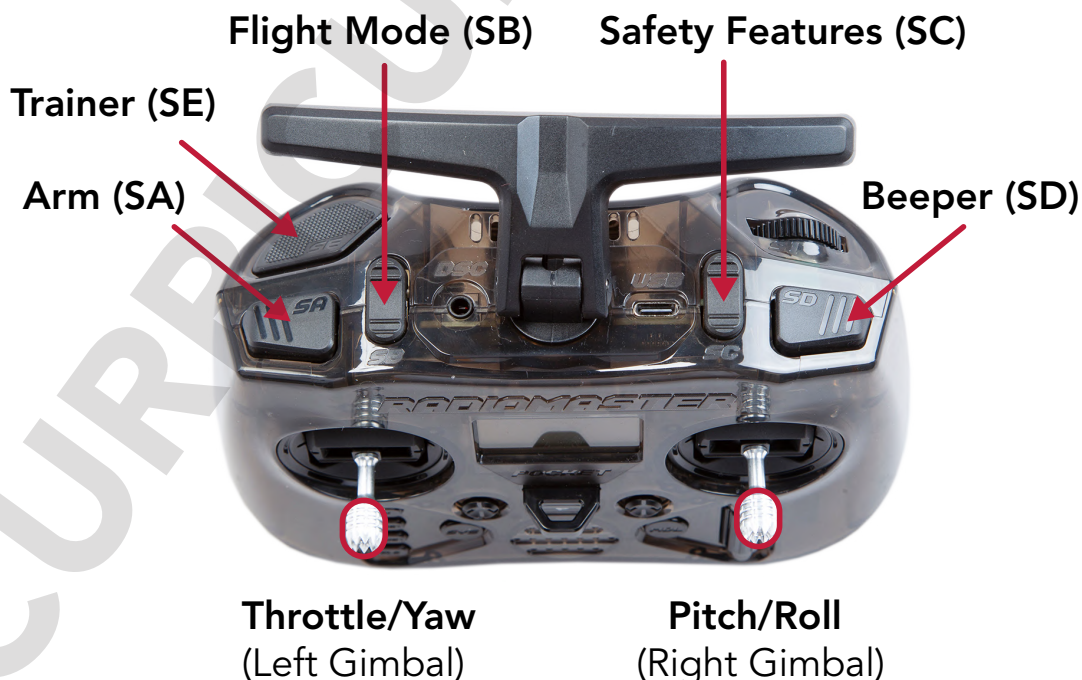
Welcome to the FLEX UAV radio system! This guide will get you comfortable with the **RadioMaster Pocket transmitter**—the radio you'll use to fly FLEX UAVs throughout your course.

The RadioMaster Pocket is a compact, professional-grade transmitter running **EdgeTX** firmware. This radio has all the features you need for safe, effective training, including three different trainer modes that let instructors supervise student flights at various skill levels.

What this guide covers:

- Physical controls and their functions
- How to charge and care for your radio
- Overview of six pre-configured FLEX models
- Trainer modes and when to use each one
- How to set up tethered (wired) training sessions

HARDWARE OVERVIEW



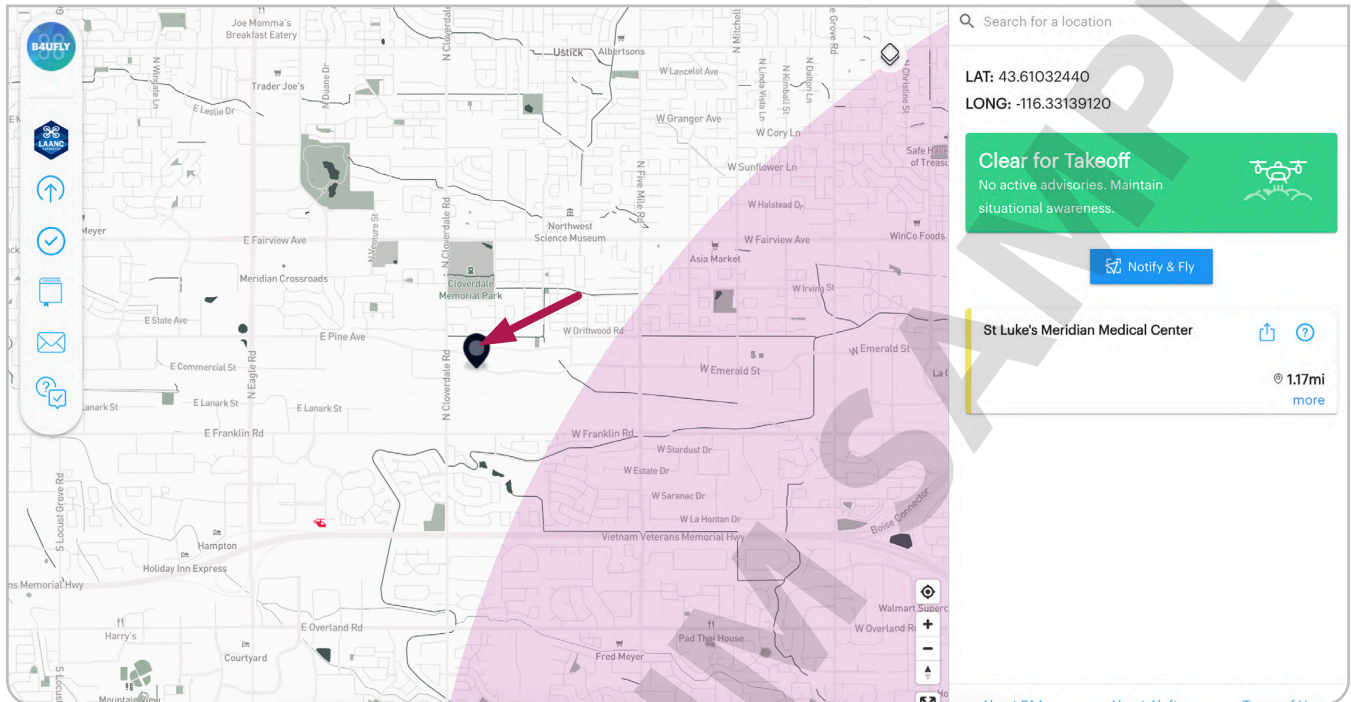


Gimbals (Sticks)

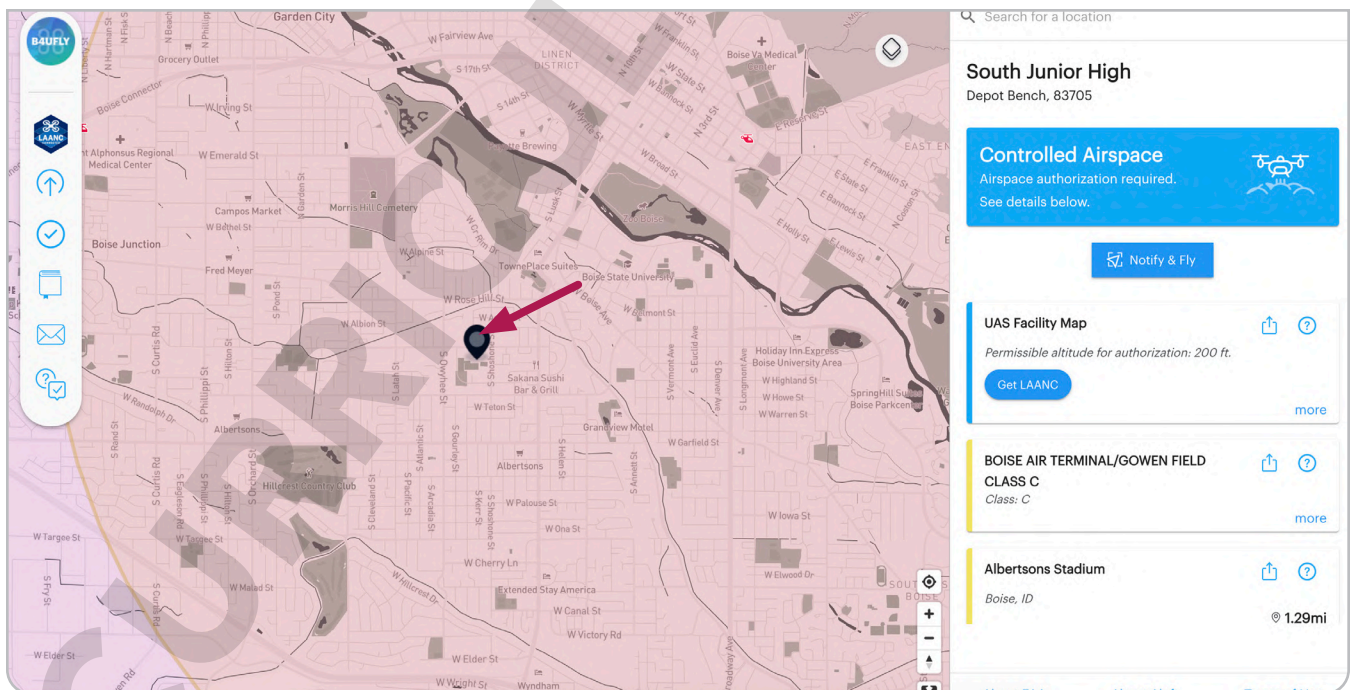
Gimbal	Controls	Spring Return
Left Gimbal	Throttle (up/down) and Yaw (left/right)	Yaw only (throttle stays)
Right Gimbal	Pitch (up/down) and Roll (left/right)	Both axes center

How to Determine Airspace Authorization

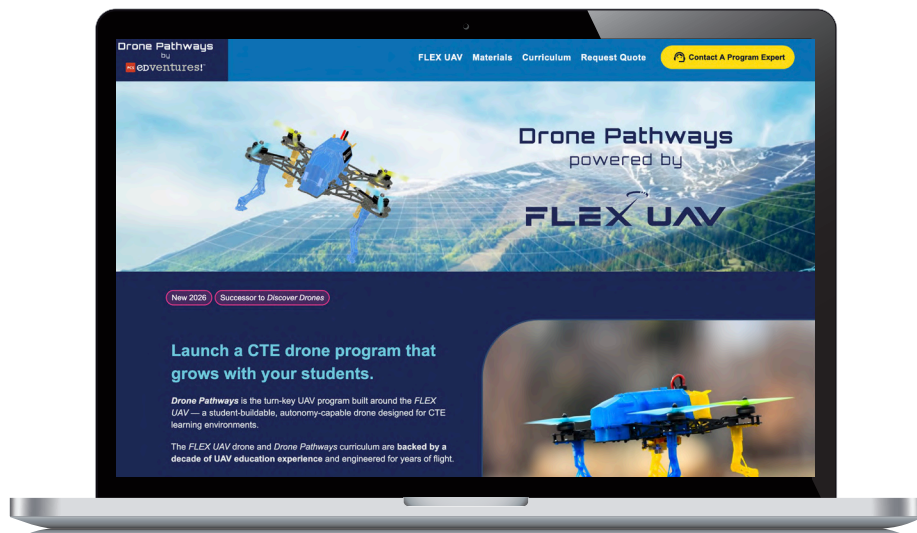
- To find out if your proposed FRIA location is in controlled airspace, visit <https://airaware.aloft.ai/>. If it does not automatically load, enter the address of the proposed FRIA location into the search bar in the top right. If you see that you are Clear for Takeoff (as shown in the first image below), no airspace authorization is required. If you are in controlled airspace (as shown in the second image below), airspace authorization is required.



NO—Airspace authorization is not required



YES—Airspace authorization is required



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Explore the **Drone Pathways** ecosystem, view the hardware in action and map out your next steps.

